

FIGURE 1.5 The term *prenatal* is created by joining the suffix **-al** to the word root **nat** and then adding the prefix **pre-**.

PREFIXES

A prefix is added to the beginning of a word to influence the meaning of that term. Prefixes usually, *but not always*, indicate location, time, or number. See Table 1.6 for a list of prefixes describing direction, quantity, size, and amount. The term **natal** (**NAY**-tal) means pertaining to birth (**nat** means birth, and **-al** means pertaining to). The following examples show how prefixes change the meaning of this term (Figures 1.5–1.8).

- **Prenatal** (pre-**NAY**-tal) means the time and events before birth (**pre-** means before, **nat** means birth, and **-al** means pertaining to).
- **Perinatal** (pehr-ih-**NAY**-tal) refers to the time and events surrounding birth (**peri-** means surrounding, **nat** means birth, and **-al** means pertaining to). This is the time just before, during, and just after birth.
- **Postnatal** (pohst-**NAY**-tal) refers to the time and events after birth (**post-** means after, **nat** means birth, and **-al** means pertaining to).

TABLE 1.6

Prefixes Describing Direction, Quantity, Size, and Amount

ab- away from, negative, absent	ad- toward, to, in the direction of
dextr/o right side	sinistr/o left side
ex- out of, outside, away from	in- in, into, not, without
macro- large, abnormal size, or long	micr/o, micro- small
mega-, megal/o large, great	olig/o scanty, few
pre- before	post- after, behind



FIGURE 1.6 The *prenatal* development of a fetus (baby).



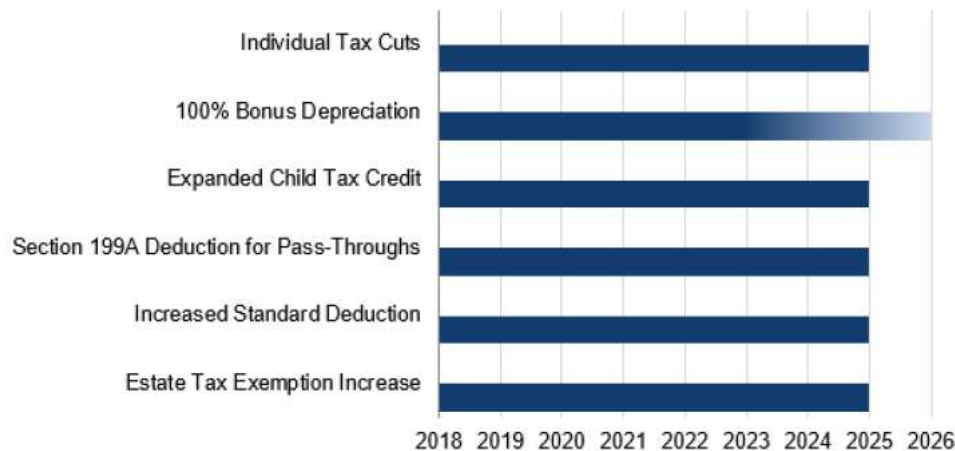
FIGURE 1.7 A *perinatal* event of the umbilical cord being cut immediately after the baby is born.



FIGURE 1.8 A joyful *postnatal* moment as the parents bond with their new baby.



Figure 2. Key Expiring Provisions of the Tax Cuts and Jobs Act



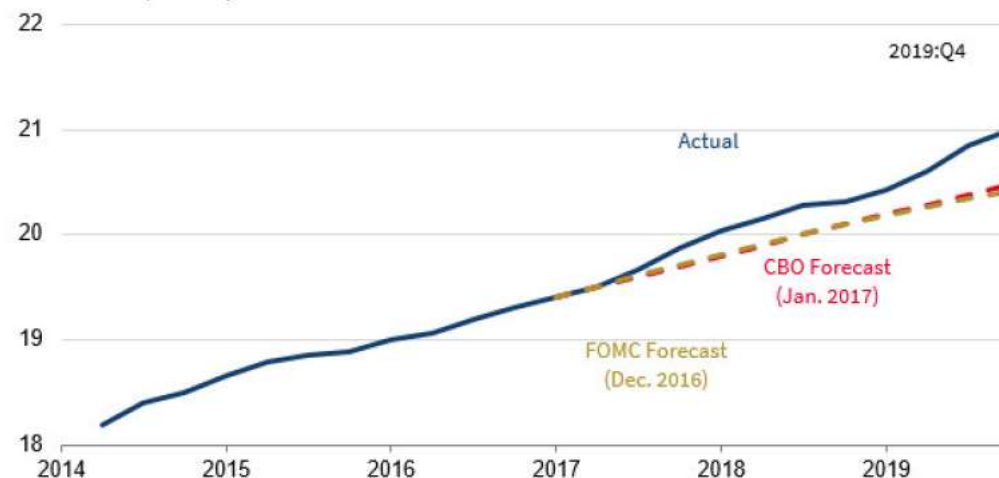
Source: Public Law 115-97

Higher Economic Growth, Lower Unemployment, and the Trump Blue-Collar Boom

Anchored by the TCJA, the economic policies of the first Trump Administration yielded higher economic growth and lower unemployment rates than projected by the CBO in January 2017 and by the Federal Open Market Committee (FOMC) in December 2016, as displayed in figure 3. The policy-induced expansion of economic supply also helped keep inflation low and stable near the Federal Reserve's 2 percent target.

Figure 3-1. The U.S. Economy Outperformed Forecasts: Real GDP

2017 dollars (trillions)



Sources: Bureau of Economic Analysis, Federal Reserve Board Federal Open Market Committee, Congressional Budget Office.

Order #: 984



Pages in Order: 1 of 1

Containers in Order: 1

REPORT TO:

65

ADRIAN
16 E 5TH ST
ADRIAN, MO 64720

BILL TO:

82

MO DEPARTMENT OF NATURAL RESOURCES
1101 RIVERSIDE DRIVE
JEFFERSON CITY, MO 65102Environmental
Sample Collection Form

Requested Analyses/Tests

PUBLIC DRINKING WATER BACTERIAL ANALYSIS

Total Coliform Bacteria and E. coli (Present/Absent Test)

PRINT LEGIBLY. Instructions for completing form are supplied in the Collection Kit. For compliance monitoring questions, contact the Missouri Department of Natural Resources-Public Drinking Water Branch at (573) 751-5331 or your regional office. For laboratory test results or testing questions, contact the Missouri State Public Health Laboratory at (573) 751-3334.

Complete or correct the following information

Collected Date: _____

yyyy-mm-dd

Collected Time: _____

24 hour format hh:mm

PWS Id: MO1010001

MO#####

Facility Id: DS

Sample Type: _____

routine, repeat, special, replacement, source

Sample Collection Point

Id: sampling point id from sample site plan

Location: _____

address or name of sampling point

Collector: _____

last name, first name

Collector Phone: _____

000/111-2222

Sample Category: Bacterial

Repeat Location: _____

upstream, downstream, original, source, other

Bottle Number: _____

Free Chlorine: _____

mg/L

Total Chlorine: _____

mg/L

Collector Signature: _____

I attest the information provided is accurate.

County: BATES

Missouri County

For Laboratory Use Only -- Please do not write below this line

Received By: _____ pH: _____

Evidence of Tampering: ☐ Yes ☐ NoEvidence of Cooling: ☐ Yes ☐ No

Date Printed: 2015-11-06

Temperature (Celsius): _____

Bottles Received: _____ Thermometer ID: _____

BUILD ID _____

BUILD ID _____

PLACE THE ACCESSION
LABEL WITHIN THIS BOX

Missouri Department of Health & Senior Services

State Public Health Laboratory

101 N. Chestnut

PO Box 570

Jefferson City, MO 65102

<http://www.health.mo.gov/lab/index.php>

Bill Whitmar, Laboratory Director

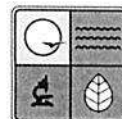


SD 062015



SAMPLE COLLECTION INSTRUCTIONS

PUBLIC DRINKING WATER for COLIFORM BACTERIA ANALYSIS



MISSOURI
DEPARTMENT OF
NATURAL RESOURCES

This sample kit and collection method is for public drinking water regulatory compliance and special samples. Only samples collected in bottles supplied by the Missouri State Public Health Laboratory (MSPHL) and collected in accordance with these instructions will be accepted for testing. **PLEASE READ THESE INSTRUCTIONS COMPLETELY BEFORE COLLECTING SAMPLES.**

Sample Containers:

Sample bottles from the MSPHL contain a chlorine neutralizer that is present in powder or liquid form. The bottles are sterile and ready for use when shipped. **Do not rinse the contents from the container and keep the bottle closed until it is to be filled.**

Shrink Wrap Seal:

Remove the seal by pulling down on the red strip and peeling shrink wrap from both the cap and bottle. Discard all shrink wrap. Do not attempt to reseal lid with shrink wrap still attached.

Two Fill Lines:

Fill the bottle until the water sample level is

BETWEEN THE TWO LINES. Place the bottle on a level surface to check the sample level.

Samples below the 100 mL (lower) line WILL NOT BE TESTED due to insufficient sample volume. Samples above the 120 mL (upper) line WILL NOT BE TESTED due to overfilled bottle. Technical protocol and EPA requirements dictate that bottles must have sufficient air space to add testing reagents and to mix the sample properly.

If the bottle is overfilled past the 120 mL line, pour off water until the sample volume is between the two lines before shipping to MSPHL. MSPHL **WILL NOT** adjust sample volume once the sample is received at the lab.

No Paper Label:

There is no longer a label to record sample information on the bottle. **DO NOT WRITE ON THE BOTTLE.** Please complete a sample information form for each sample submitted for testing. **DATE AND TIME OF SAMPLE COLLECTION and the BOTTLE NUMBER (from sticker on bottle) ARE REQUIRED.** A form for each bottle is included in this sample kit.



For More Information, please contact:

Missouri Department of Health and Senior Services
State Public Health Laboratory
Environmental Bacteriology Unit
101 North Chestnut St., P.O. Box 570
Jefferson City, MO 65102

Phone: 573-751-3334

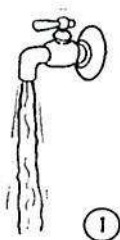
FAX: 573-522-4032

Email: labweb1@health.mo.gov

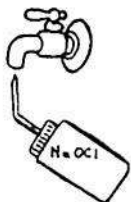
Website: www.health.mo.gov/Lab

Bacteriological Sample Collection Procedures

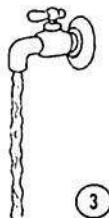
Assemble all of the sampling supplies. Before you begin, wash your hands thoroughly before handling supplies. Go to the sampling location(s) specified in your Missouri Department of Natural Resources (MDNR) approved sampling site plan. The sample should be taken from a clean, smooth-nosed cold water faucet if possible. Avoid drinking fountains, leaky faucets, hot/cold mixing faucets and frost-proof yard hydrants since it is not practical to sterilize these fixtures. If possible, remove any aerators, strainers or hoses that are present because they may harbor bacteria. Follow the procedures below when collecting the sample. Instructions for completing the environmental sampling form are on the following page.



①

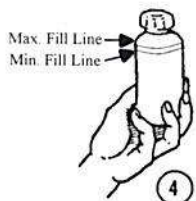


②



③

Bottle is clean and sterile,
ready for sample
(Remove and discard plastic seal)



④



⑤

1. **Open the cold water tap for about 3 minutes before collecting the sample.** This should adequately flush the water line of any debris.
2. **Flame-sterilize the tap and/or chemically disinfect the tap. Do not flame-sterilize if tap is plastic or if aerators are attached.** Disinfect tap by thoroughly rinsing both the inside and outside of the tap with a mixture of 50% house-hold bleach (NaOCl) and 50% tap water. Take extreme care with strong bleach (oxidizing) solutions.
3. **Flush the tap for an additional 3 minutes with cold water, and then reduce to a gentle flow to about the width of a pencil.** Do not change the water flow once you have started sampling as this could dislodge contaminants in the tap.
4. **Remove the plastic shrink wrap seal by pulling down on the red strip and peeling the shrink wrap from both the cap and bottle. Discard the shrink wrap.** Do not attempt to reseal the lid with shrink wrap still attached.
5. **Grasp cap along top edge and remove carefully. Do not touch the inside with your fingers.** Hold the bottle in one hand and the cap in the other. **Do not lay the cap down or put it in a pocket!** Also, take care not to contaminate the sterile bottle or cap with your fingers or permit the faucet to touch the inside of the bottle.
6. Hold the bottle so that water entering the bottle will not come in contact with your hands or the outside of the bottle.
7. **Fill the bottle until the water sample level is BETWEEN THE TWO LINES on the bottle (100 – 120 ml).** Preferably, the sample level should be at or just slightly above the 100 ml line. **Sample levels below the 100 ml (lower) line WILL NOT BE TESTED due to insufficient sample volume. Sample levels above the 120 ml (upper) line WILL NOT BE TESTED due to overfilled bottle.** If the bottle is overfilled, you may pour off any excess water to get the sample level between the two lines. Place the cap on the bottle and screw it down tightly.
8. **Fill out the Missouri Department of Health and Senior Services (DHSS) State Public Health Lab (SPHL) Environmental Sample Collection Form using waterproof ink. See attached document for instructions on properly completing the sample collection form and for shipping instructions.**
9. For single samples, neatly fold the sample collection form into thirds (standard letter fold), roll around the bottle and place in the shipping box. For multiple samples, fold the forms once or more as needed, and place in the shipping box alongside the samples. If needed, use bubble pack or folded paper to fill space. Do not use shredded paper. Seal the box with a single strip of shipping tape and affix the return address label to the top of the box.



⑥

Fill sample bottle until water level is
between the two lines on the bottle.



⑦



DO NOT USE.



DO NOT USE.

Contract operators will be provided with forms for all the supplies they operate. Blank forms will be available for MDNR Regional Office staff use.

3. The form requires all requested information to be printed by the collector. There are no longer check boxes for **Sample Type** or **Repeat Location**.
4. **Facility ID, Sample Collection Point ID and Location** for the sampling site **MUST** be provided by the collector. This information is available from your MDNR approved PWS sampling plan. MDNR will be providing all public water systems with a current copy of their approved sampling plan. This information is required by SDWIS and is used by MDNR to ensure regulatory compliance requirements have been met. Failure to complete this information on the sample collection form may result in a non-compliance report from MDNR.
5. A **Collector Signature** line has been added. The sample collector must sign the form to attest the information provided is accurate to the best of their knowledge.

The MSPHL will begin shipping the new forms to public water systems in late November or early December. **Please begin using the new forms December 16, 2015.** Discard all the old forms (“cards”) at that time.

NEW SAMPLE INSTRUCTIONS:

Sample instructions have been revised to include changes to the bottle and sampling form. The instructions include detailed information on how to collect the sample using the new bottle, how to complete the new sample collection form, how to best ship samples to the MSPHL using the free MSPHL courier system, and how to register for the new MSPHL web portal. A copy of these instructions is attached.

NEW WEB PORTAL FOR RESULTS REPORTS

The OE LIMS provides a web portal that may be used by systems to view and print their test result reports, check status of samples, download sample information into Excel, and receive automated emails when samples are received at the laboratory, and when sample results are ready to be viewed. For information on how to gain access to this portal, please contact Shondra Johnson, LIMS Administrator at Shondra.Johnson@health.mo.gov or at 573-751-3334.

IMPLEMENTATION DATES:

The MSPHL intends to implement the OpenELIS LIMS on **December 1, 2015**. There will be a two week testing period in which laboratory staff will run the new LIMS in conjunction with our current manual, paper-based system to ensure the OE LIMS is operating properly. You may continue to submit samples as you currently do, using the old sample information card, throughout this time.

On December 16, 2015, the MSPHL plans to “go-live” with the new OE LIMS. Samples submitted after that date should be submitted on the new Environmental Sample Collection Form. At that time, the MSPHL Test Results Web Portal will also be available to those systems that have been granted access.

The MSPHL and MDNR understand that there will be a lot of changes to a system that has been in place for many years. The MSPHL is excited about the added benefits from this new system, and we ask for your patience as we implement the OpenELIS LIMS at the Missouri State Public Health Laboratory.